HUMAN-COMPUTER INTERACTION · INF13186L

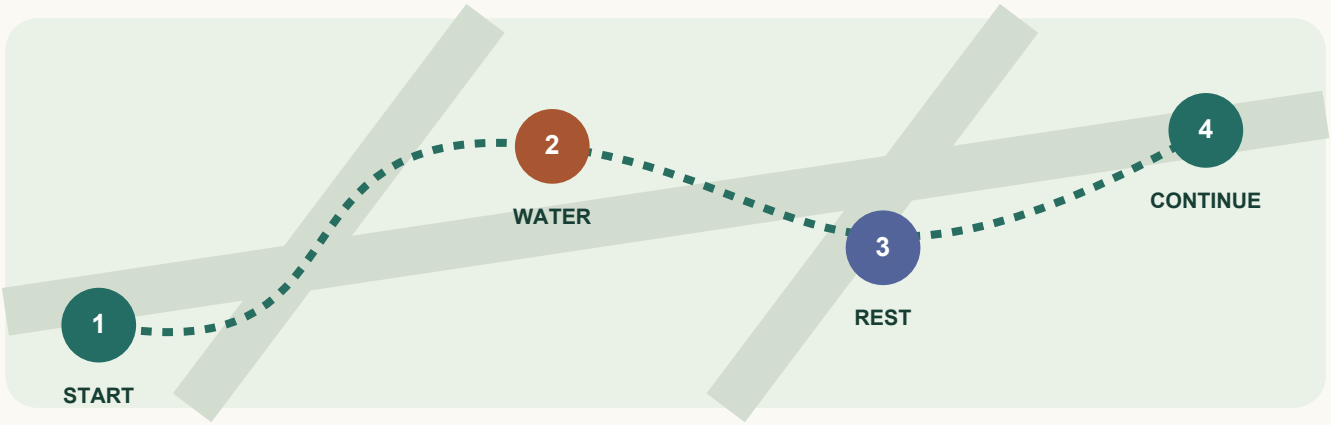
AlunUÉ
Water along the way

Exploratory research and requirements for refill discovery and student walking journeys in Évora.

Eren Ozturk · University of Évora · 30 October 2026

English review copy. The Portuguese PDF is the intended coursework version. This copy explains the same evidence, design proposals and limitations for the author to review.

Open the app · GitHub Pages | Repository and checks



Conceptual journey, not a street map or verified directory.

Scope and course fit

The proposal is an essential-services and mobility module within AlunUÉ: find trustworthy information about a place to refill a bottle and continue the student day. Supermarkets, cultural stops and willing shops are optional extensions. Phase 1 focuses on research and requirements; a working prototype supports exploration and does not establish that the concept has been validated.

State of the work

This is individual work. The author says the lecturer approved individual work; documentary approval is not presented. Available research includes three individually reconstructed conversations from the previous week and general recalled feedback. The three personas and maps are provisional and now trace to concrete needs in those accounts. No recoverable structured survey or usability study has been completed. Evidence, interpretation and proposed design are kept distinct; missing data is not replaced by fabricated findings.

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Method and exploratory signals

Source and conditions of collection

The author informally spoke with acquaintances, including Erasmus students and people in his faculty, during the week before his account on 30 October 2026. Following loss of the original notes, he reconstructed three individual cases: C1-C3 (appendix). The author confirmed that all three are UÉ students and agreed to anonymous coursework summaries. Exact dates and locations were not recovered; enrolment and permission are confirmed through the author's account. About 30 interlocutors remains an approximate recollection; the three cases are not a representative sample.

Remembered prompts asked what people would like and whether an app would be useful. The proposal was discussed, but the precise question order was not recovered. This is retrospective reconstruction from a convenience sample, subject to selective recall and social-desirability bias. It does not measure adoption or problem frequency.

ID / source	Available signal and implication
E1/E2 · general recollection	Positive reactions, repeated water purchases and lack of awareness of refill locations, according to the author. No percentages or auditable counts.
E3 · author preference	Trust, walking routes, groceries, views and voluntary shops are scope decisions, not survey rankings.
E4 · operational state	No shop is confirmed as a participant. A network requires future agreements.
E5 · C1	Difficulty finding fountains in a new city and perceived difficulty finding inexpensive purchases and card payment. Investigate discovery, cost and payment information.
E6 · C2	A regular runner wants places to drink water and rest. Investigate route continuity and rest stops.
E7 · C3	Positive reaction and interest in receiving news. Investigate voluntary updates; this does not establish adoption or authorise automated notifications.

Interpretation and participant protection

The cases support specific needs to investigate, not generalisations about all students. C1's view of local retail is a perception, not a survey of Évora businesses. No individual objections were recalled; this does not establish their absence. Anonymous codes replace names; contacts and private messages are not published. The author confirmed permission for anonymous summaries of these three cases. Obtain renewed permission if the use changes.

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Three personas traced to the accounts

These are provisional need-based personas anchored in C1-C3, not invented biographies or statistically validated segments. The author confirmed all three interlocutors' UÉ connection. Generalising each profile to other students still requires validation. No unreported age, nationality, income or occupation is assigned.

P1 · Discovery in a new city / C1

Goal: find water and inexpensive alternatives without knowing the city. Recalled issues: fountains are hard to find; uncertainty about low-cost purchases and card acceptance. Implication: clear sources and cost; show payment information only when verified. Do not assume language ability.

P2 · Runner with water stops / C2

Goal: drink water and rest along regular running routes. Recalled fact: runs often and wants water/rest locations. Implication: plan and verify stops. Running times, distances and medical needs were not collected.

P3 · Person interested in following the service / C3

Goal: learn when the service is available. Recalled response: positive reaction and interest in news. Implication: a future voluntary update mechanism. Preferred channel, frequency, notification tolerance and actual usage remain unknown.

Supported needs and remaining validation

Discovery and cost trace to C1; running, water and rest to C2; interest in news to C3. Emotions, detailed flows and interfaces are design proposals. Each profile has only one reconstructed source and needs further interviews to establish recurring patterns. P3 describes communication interest, without establishing a third distinct refill need.

Next research: ask for concrete experiences before showing the prototype, investigate water/rest and update preferences, preserve negative responses and record date, context and permission. Do not replace memory gaps with assumptions.

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Empathy maps with traceability

“Says / goal” summarises recalled feedback without quotation marks or claims of verbatim transcription. Other dimensions distinguish reported facts from hypotheses.

P1 · Discovery in a new city / C1

SAYS / GOAL

Recalled: fountains are hard to find; affordable purchases and card payment matter.

THINKS

Hypothesis: how can I find an alternative and know the cost?

DOES

Not reconstructed: actual search route, purchases or app use.

FEELS

Not collected; possible uncertainty remains a hypothesis.

P2 · Runner / C2

SAYS / GOAL

Recalled: wants places to drink water regularly and rest.

THINKS

Hypothesis: can I fit water and rest into the route?

DOES

Reported: runs often. Routes, distances and times unknown.

FEELS

Not collected; do not assume fatigue or anxiety.

P3 · Following the service / C3

SAYS / GOAL

Recalled: positive about the idea and would like news.

THINKS

Hypothesis: how will I know when the service is available?

DOES

Expressed interest in updates; product use was not observed.

FEELS

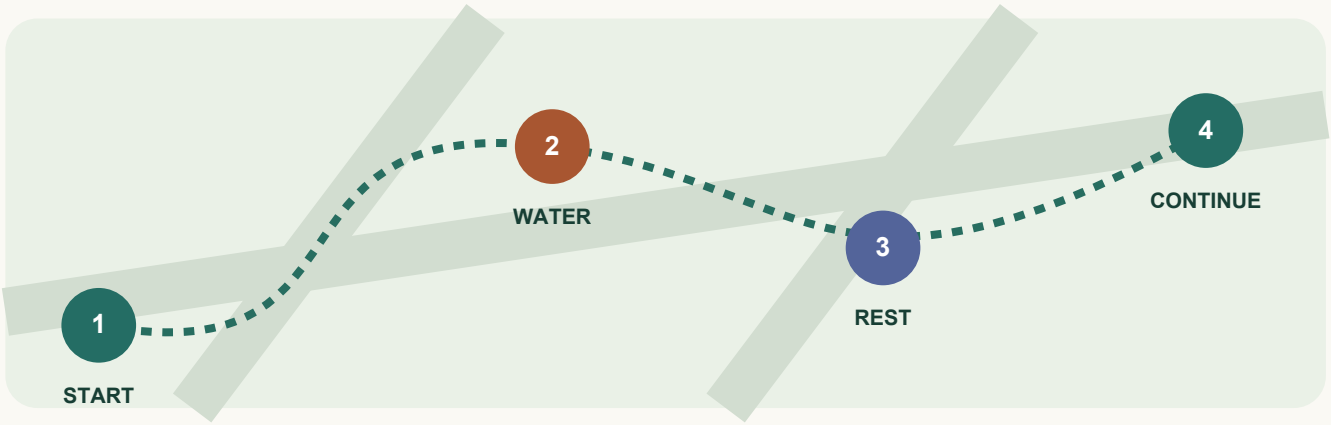
Positive reaction recalled; intensity not measured.

Confirm interpretations with the people concerned. Empathy maps do not establish direct knowledge of their thoughts.

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Proposed user journey

Design scenario: a student needs to refill a bottle while travelling through the city. This sequence has not been observed in a field study or usability test.



Conceptual journey, not a street map or verified directory.

Stage	Action and possible difficulty	Design response
1 · Need	Look for water without knowing where to start.	Searchable list, optional location and manual start.
2 · Assess	Distinguish a useful point from uncertain map data.	Source, date, access, cost and verification status.
3 · Plan	Choose stops without an unnecessary detour.	Adjustable order and optional grocery/cultural stops.
4 · Travel	Move the plan to external navigation.	Walking directions export and a text alternative.
5 · Arrive	Encounter a closure or incorrect information.	Show limits, seek an alternative and prepare a correction.
6 · Continue	Continue without pressure to spend or interact.	No purchase requirement; shops only with explicit opt-in.

Variations traced to the accounts

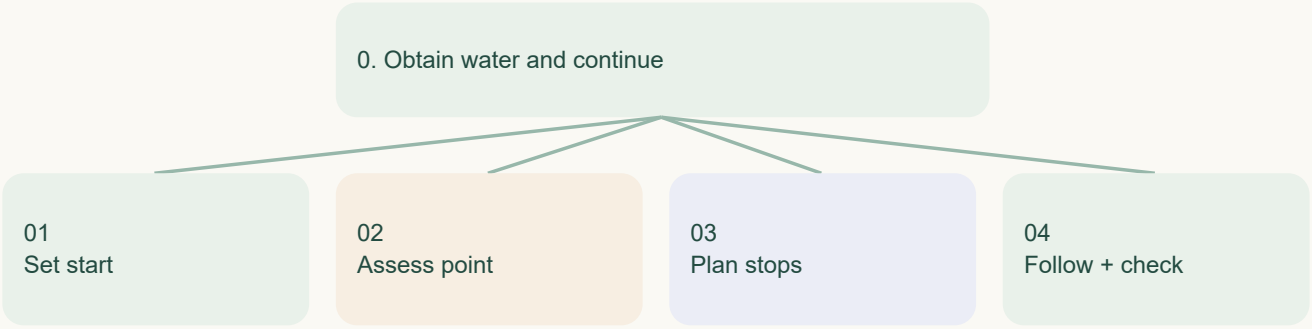
P1 / C1: discover water and inspect verified cost/payment information. P2 / C2: plan water and rest while running. P3 / C3: discover proposal → request news → choose channel and consent → receive news → opt out. The update flow is proposed and not implemented.

The diagram is not a street map or verified directory. OpenStreetMap candidates require checks of potability, free access, availability and operation.

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Hierarchical task analysis

Goal 0: obtain water and continue the journey. This HTA describes intended tasks and recovery paths, rather than behaviour already observed.



Plan: 1 → 2 → 3 → 4. If verification fails, choose another point. If export fails, save text.

Task	Subtasks and completion conditions
1 · Set a start	1.1 Choose centre/campus or coordinates. 1.2 Request location only if wanted. 1.3 Keep a manual option when permission is refused.
2 · Assess a point	2.1 Inspect the list. 2.2 Read source and date. 2.3 Check water suitability, cost, access and operation. 2.4 Reject unconfirmed candidates from a real water journey.
3 · Plan stops	3.1 Add an eligible point. 3.2 Optionally add a supermarket or cultural stop. 3.3 Reorder. 3.4 Review. 3.5 Export in parts or save text.
4 · Follow and check	4.1 Consult external navigation. 4.2 Confirm conditions at the location. 4.3 Refill if allowed. 4.4 Choose an alternative and record a suggestion if information fails.

Plans and exceptions

Main plan: 1 → 2 → 3 → 4. Repeat task 2 if no eligible point is available. Without internet, show a dated stored snapshot and saved itinerary; do not promise offline street maps. Without location, use task 1.1. If external services fail, preserve the list and offer text. A shop without agreement must not be shown as a participating partner.

Operations explored in the prototype

Type filters, search, an identified demonstration mode, selected and ordered stops, walking export, share/copy with text-file fallback, device-only suggestion drafts and optional service requests. There is no verified operational directory, moderation service or real partner registration.

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Requirements and evaluation criteria

Project priorities: P0 essential; P1 extension. Targets below are proposed criteria for future testing, not measured results.

ID / priority	Requirement and traceability	Proposed criterion
R1 · P0	Distinguish candidates, demonstrations and confirmed points. E2 plus misinformation risk.	Never describe an unverified candidate as guaranteed drinking water.
R2 · P0	Find water without an account or mandatory location. E1/E2 and accessibility hypothesis.	At least 4 of 5 participants complete a manual-start task.
R3 · P0	Show source, date, cost and access. E2 and trust decision.	At least 4 of 5 correctly identify uncertainty before choosing.
R4 · P0	Preserve all stops on export. E3.	No chosen stop is lost; a readable text alternative exists.
R5 · P0	Recover from network/API failure. Robustness decision.	The plan is preserved and the failure explained.
R6 · P1	Combine optional grocery/cultural stops. E3.	Each extension can be excluded while keeping water as the focus.
R7 · P1	Voluntary shops without mandatory purchase. E3/E4.	No shop is listed as a partner without explicit agreement.
R8 · P1	Water/rest along a route; cost/payment when verified. E5/E6.	Never claim confirmed rest, price or card acceptance without a source.
R9 · P1	Future updates with explicit opt-in. E7.	No contact collection or notifications without permission; allow withdrawal. Not implemented.

Usability and accessibility

Design goals: European Portuguese labels, visible focus, keyboard access, readable contrast, zoom without content loss, suitable touch controls and recovery messages. Evaluate status recognition, completion, errors, task time and satisfaction. No SUS result was collected in this phase.

Information architecture

Discover: search, filters and sources. My journey: start, stops and export. Community: opt-in proposal and local drafts. The project: evidence, limits and sources. Planning a journey does not require joining the shop network.

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Limits, next steps and sources

Limits affecting this phase

The research is exploratory and retrospective. Three individual accounts have been reconstructed, but original notes/transcripts, a structured questionnaire, a verified participant population and independent persona validation are missing. A positive reaction to a presented idea does not measure adoption. The design artefacts must therefore not be treated as the results of a completed user study. Losing the notes explains the limitation but does not remove it.

Step	Outcome to obtain
1 · Further research	Neutral interviews with real students; establish needs before showing the concept and preserve authorised anonymous summaries.
2 · Revise Phase 1	Update personas, maps and requirements from collected evidence; keep a change record.
3 · Evaluate prototypes	Paper and medium-fidelity tasks, followed by heuristic evaluation required by the course.
4 · Final phase	At least five real UÉ users, task metrics and SUS; report only actual observations.
5 · Operational service	Verify water points in the field, assign maintenance and obtain shop agreement. Test the Android PWA before deciding on APK distribution.

Use of generative AI

Generative AI assisted report structure and Portuguese drafting, design artefacts, diagrams, prototype code and translation. The author supplied recollections and scope decisions. AI did not conduct interviews or verify places. The author should review and understand the content and remains responsible for the authenticity of the submission.

Sources and supporting materials

1. INF13186L, Introdução à UC, pp. 5-6: AlunUÉ scope, ten-page report limit and AI disclosure.
2. INF13186L, gemini_fase1, p. 1: research, personas, empathy/journey maps and HTA. Later example pages were not used as interview evidence.
3. Author statements and recollections, 30 October 2026.
4. OpenStreetMap attribution; official Google Maps URLs documentation.
5. Public app and dedicated repository. The PDF stands on its own; links do not replace the required design artefacts.

Appendix A · Reconstructed accounts

Source: the author's account on 30 October 2026, describing conversations during the previous week. The author confirmed all three people's UÉ connection and agreement to anonymous coursework summaries. Anonymous codes replace names. Questions and answers are paraphrases, not direct quotations. Exact dates, locations and question order were not recovered.

C1 · First person consulted

Field	Recalled summary
Question	What would you like to find, or what would be useful in this proposal? Approximate wording.
Response	Fountains are hard to find, and it can be difficult to know what to do in a new city. Mentioned difficulty finding inexpensive items such as water and shops accepting cards, and a perceived lack of chain shops.
Interpretation	Investigate discovery and cost/payment information. The retail perception does not establish actual business conditions.

C2 · Person who runs regularly

Field	Recalled summary
Question	What would you like to find in the service? Approximate wording.
Response	Runs often and wants places to drink water regularly and rest.
Interpretation	Water and rest are explicit needs. Route, exact frequency, duration and locations were not recalled. The author subsequently confirmed this person's identity; the analysis retains the anonymous code.

C3 · Person interested in news

Field	Recalled summary
Question	The author recalls asking similar questions about the proposal; the precise sequence is unavailable.
Response	Reacted positively, considered the idea useful and wanted to receive news.
Interpretation	Declared interest in following progress, with no channel or frequency established. This is neither permission to publish personal data nor proof of adoption.

The author also reports a list of other interested people, but no names, contacts or further responses were supplied. That list is not counted as additional participants or treated as authorised notification subscriptions.

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Visual synthesis · Evidence to design

Each row connects a reconstructed account to a need, a working persona and the corresponding requirements. Colours identify cases; they do not encode percentages, intensity or frequency.

ACCOUNT	RECALLED NEED	WORKING PERSONA	REQUIREMENT
C1	Discover + cost/payment	P1 · New-city discovery	R3 / R8
C2	Water + rest while running	P2 · Running with stops	R4 / R8
C3	Voluntary progress updates	P3 · Following the service	R9

How to read the project

Layer	What can be claimed
ACCOUNT	The author recalled three conversations and confirmed UÉ connection and permission for anonymous summaries.
INTERPRETATION	Themes guide design. The connections shown are project analysis, not statistical conclusions.
PROPOSAL	Personas, maps, targets and future features require confirmation through research and testing.

The validation path

Recall and document → confirm with participants → deepen the research → revise artefacts → test tasks → measure and improve. Visual clarity communicates the work; evidence quality supports its decisions.